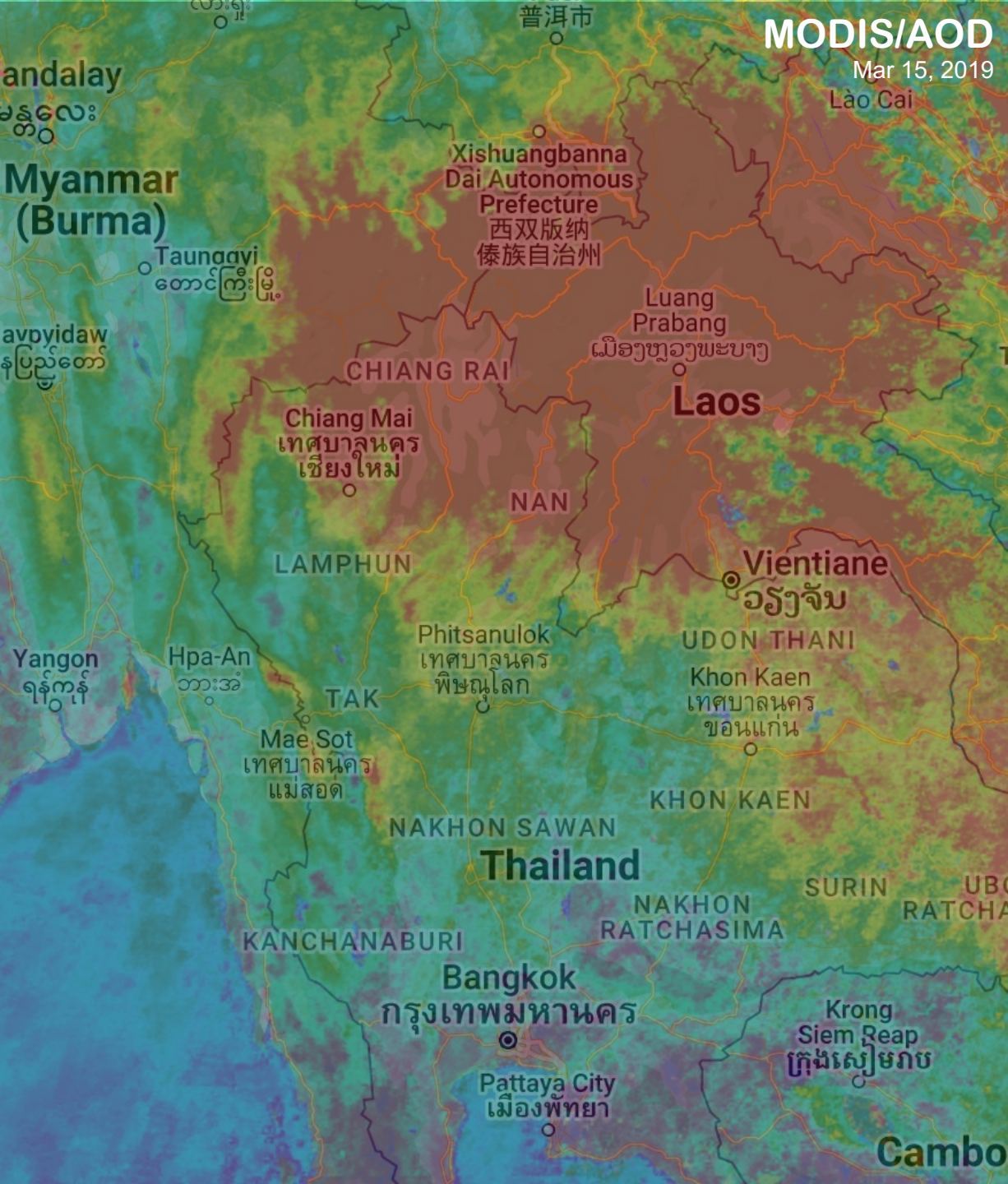




The Impact of Transboundary Haze on Lung Cancer Mortality in Thailand

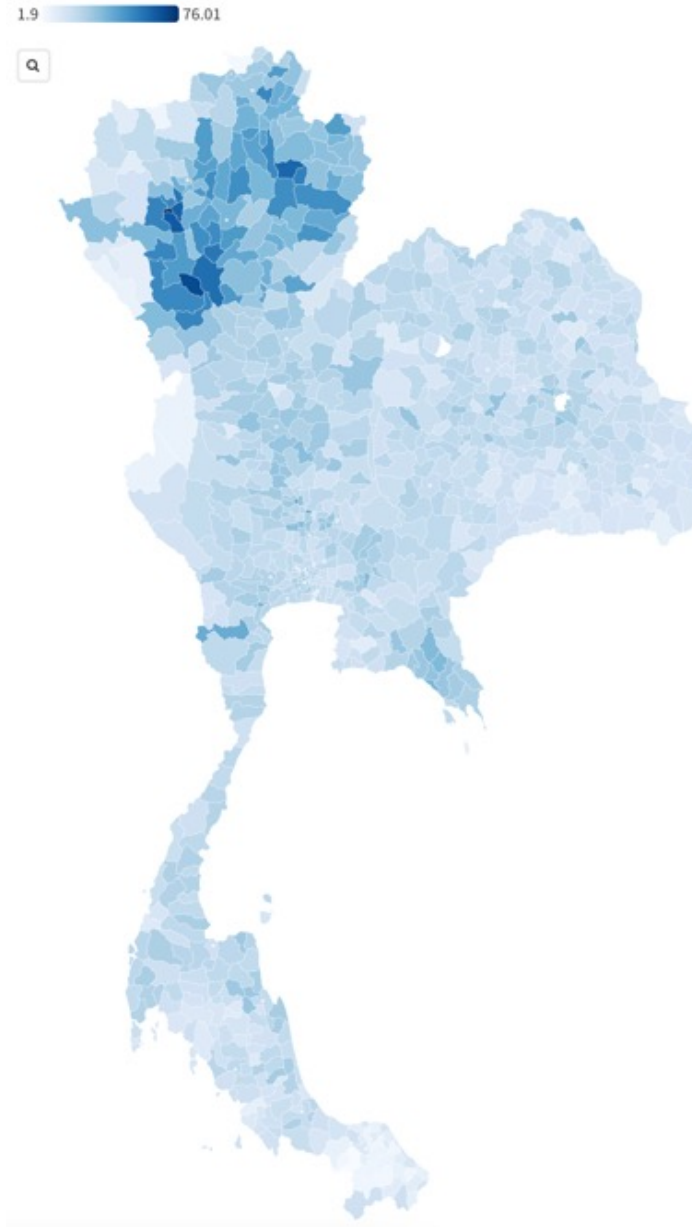
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Advisor: Assistant Professor Jun Goto

Motivation

- The data shown in *Figure 1* indicates that on average the northern area of Thailand has a greater death rate from lung cancer than any other location
- But not from the smoking (*Figure 2*), because the ratio of smoking population in the north not different from another region. So, there might be other factors.

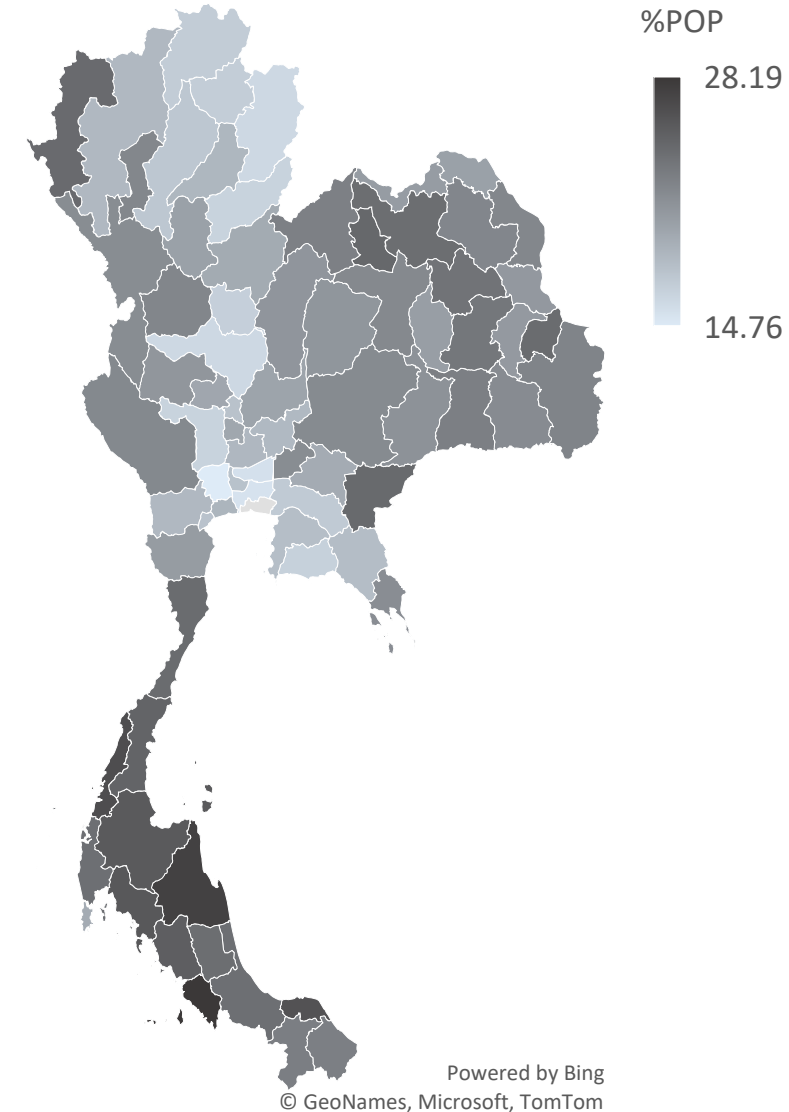
Seasonal haze might cause the air quality and link to health problems?

Figure 1: Average death rate from lung cancer per 100,000 persons in Thailand from 2016 to 2020



Source: Strategy and Planning Division, Ministry of Public Health

Figure 2: Average tobacco consumption rates aged 15 years and older in Thailand from 2011 and 2017



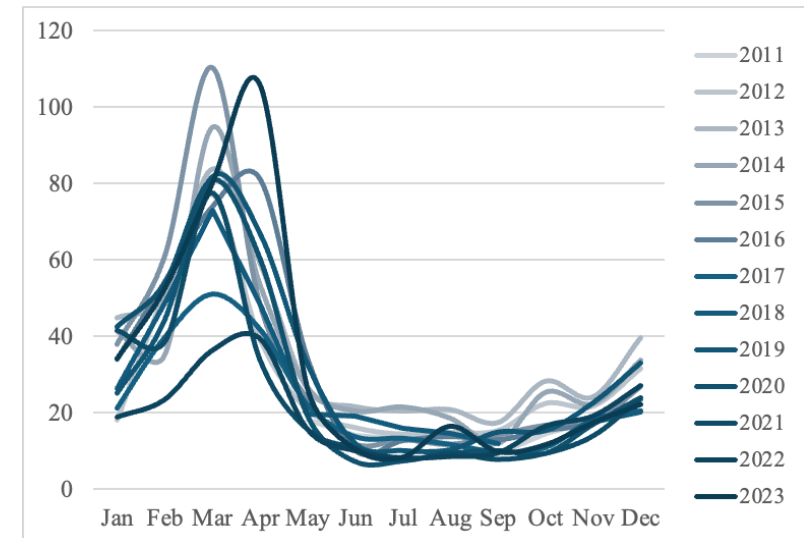
Source: National Statistical Office of Thailand

Assumption



The northern part of Southeast Asia suffers from seasonal haze during January to April, which contains harmful tiny particles including **PM_{2.5}** that impact on health.

Figure 3: Monthly average PM 2.5 ($\mu\text{g}/\text{m}^3$) in the center Chiang Mai (36T), Thailand from 2011 to 2023



Source: Pollution Control Department

an example of seasonal haze patterns in the center of Chiang Mai.

Research Questions

Q

Is there a connection between lung cancer and the effects of transboundary haze in northern Southeast Asia?

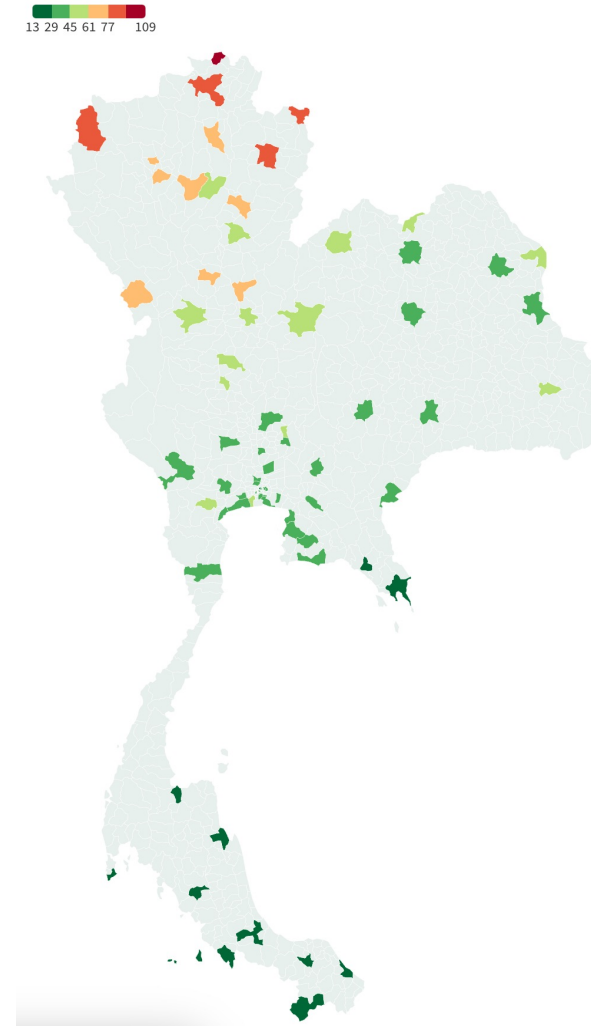
However, there are many limitations.

- 1) The lung cancer mortality at the district level **available only in Thailand**.
- 2) PM2.5 data provided by ground-based air quality measuring stations are limited in **area coverage**.
- 3) Aerosols from AOD (covered all 928 districts) are tiny particles suspended in the atmosphere **may not have the same direct impact on health as PM 2.5**.

Assumption

- However, monitoring air quality and small particles relying on **ground-based stations** has the limitations in terms of the length of historical data and its coverage. *(Figure 4)*
- There are only **61 stations** (6.6% of Thailand's total of 928 districts), and some of them have a maximum historical data period of 10 years.

Figure 4: Ground-based air quality measuring stations in Thailand (not covered)



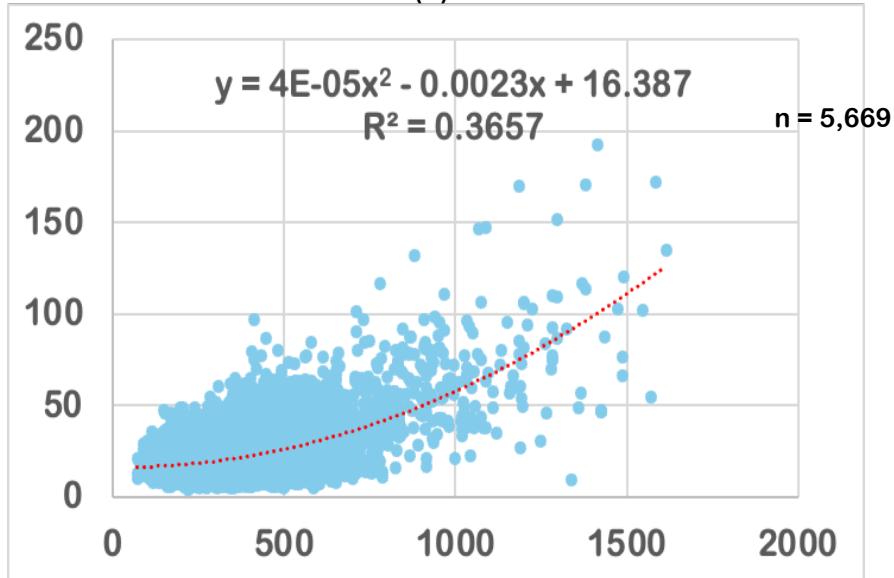
Source: Pollution Control Department

Air quality data using satellite remote sensing which offer broader coverage might be the solution.

Assumption

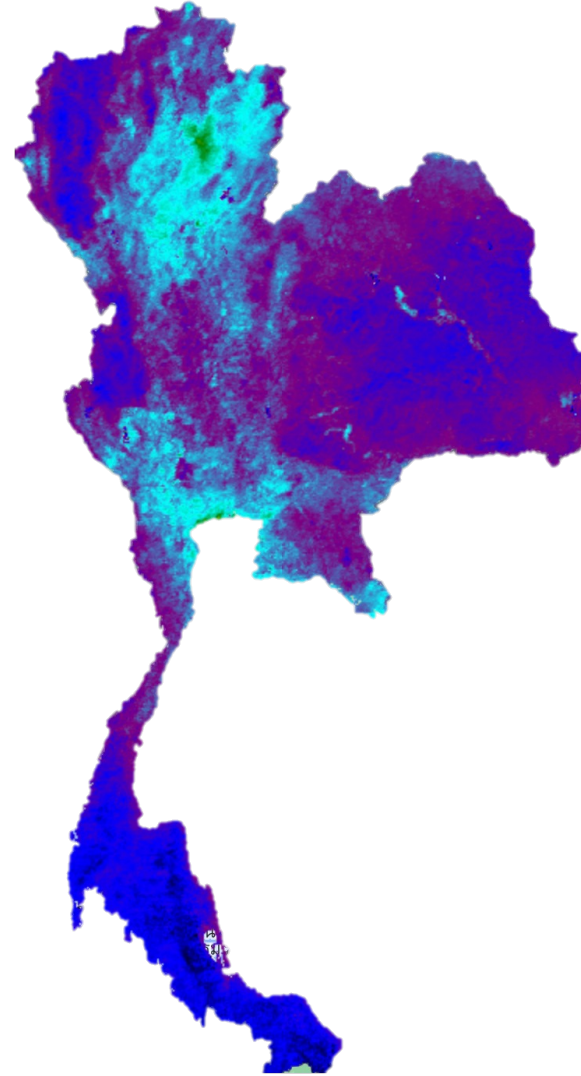
- AOD data from MODIS¹, which has been available **since 2001 covered all areas of interest.** (Figure 5)
- Figure 7 shows a **positive correlation** between The Aerosol Optical Depth (AOD) and PM2.5 levels.

Figure 7: Relationship between monthly average PM 2.5 (y) and AOD (x) in Thailand



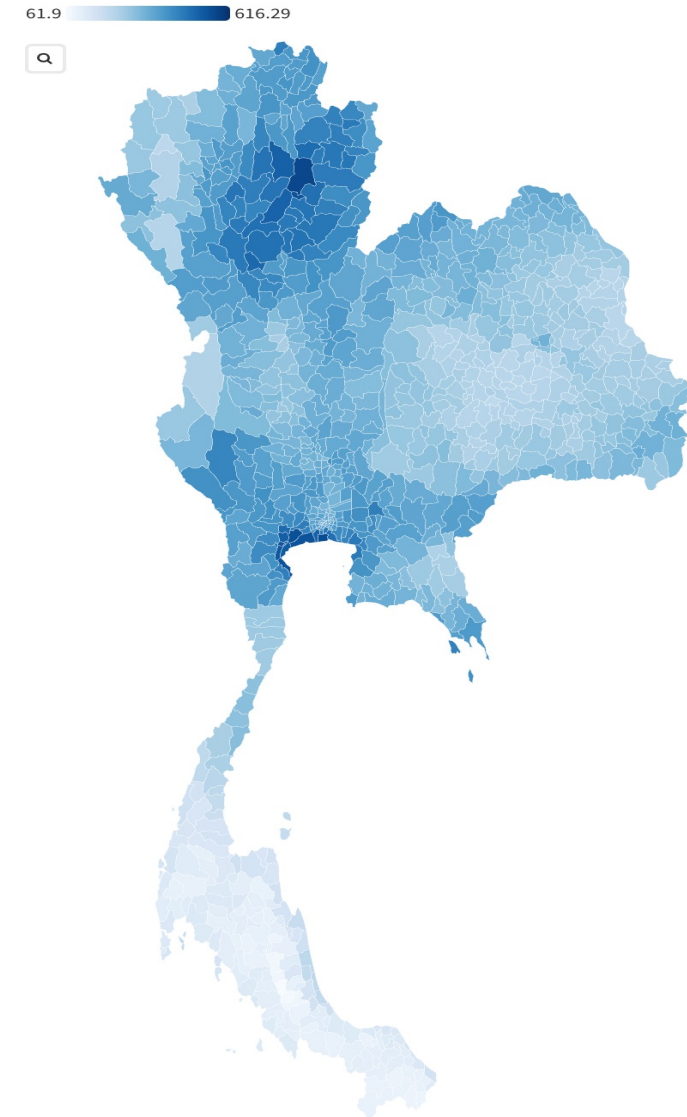
Source: Author's calculation

Figure 5: The Aerosol Optical Depth (AOD) from MODIS in Feb 2021



Source: NASA LP DAAC at the USGS EROS Center

Figure 6: The monthly average Aerosol Optical Depth (AOD) from MODIS in Feb 2021 at districts level



Source: NASA LP DAAC at the USGS EROS Center

Note: ¹MODIS (Moderate Resolution Imaging Spectroradiometer) is a sensor aboard NASA's Terra and Aqua satellites that observes Earth's surface and atmosphere.

$$\text{Lung cancer}_d = \beta_0 + \beta_1 \text{Air Quality_normal}_d + \beta_2 \text{Air Quality_haze}_d + \gamma \text{Controls}_d + \varepsilon_d$$

d = district level (ADM2)

Lung cancer_d = Average Lung cancer Mortality Rate (2017-2021)

Air Quality_normal_d = Average PM2.5 / AOD during off haze season (May-Dec)

Air Quality_haze_d = Differentiate average PM2.5 / AOD between during haze and off haze season
(PM2.5 or AOD during Jan-Apr different from May-Dec)

Controls_d = Tobacco consumption rate, Household income and other air quality (nitrogen dioxide, sulfur dioxide, carbon monoxide, and ozone)

averaging pollutant levels over extended periods (5, 10, and 20 years)

Data Collection

- On average, Northern region has higher lung cancer mortality, PM2.5, AOD than the others.
- But their tobacco consumption rates were lower than average

Table 1: Descriptive statistics

Variables	Region	Obs	Mean	SD
Lung Cancer Mortality (per 100,000 people)	TH	4,640	22.28	11.11
	Central	1,295	23.35	8.50
	North	980	30.67	15.03
	Northeast	1,610	18.88	7.47
	South	755	16.79	8.68
Particulate Matter 2.5 (µg/m3)	TH	4,897	23.76	16.13
	Central	2,621	23.92	12.17
	North	1,195	27.48	24.09
	Northeast	476	24.64	15.47
	South	605	15.05	5.12
Aerosol Optical Depth	TH	259,727	404.52	192.99
	Central	67,891	429.45	165.34
	North	54,962	443.30	228.65
	Northeast	90,115	434.63	196.13
	South	46,759	264.72	87.43
Tobacco Consumption Rates (percent)	TH	154	20.92	3.76
	Central	52	18.88	3.13
	North	34	19.27	3.48
	Northeast	40	22.25	2.58
	South	28	24.84	2.68
Household income (Thai baht)	TH	770	20,423.44	6,987.84
	Central	260	24,207.62	7,449.38
	North	170	16,775.54	4,446.61
	Northeast	200	16,824.63	4,707.09
	South	140	22,966.43	6,540.24
Elevation (meter)	TH	928	190.48	194.49
	Central	259	61.17	99.20
	North	196	396.59	273.51
	Northeast	322	206.38	94.27
	South	151	110.84	96.28

Results

Table 2: The impact of PM2.5 from ground stations for 10- and 5-years exposure on lung cancer mortality

Y = Average Lung cancer Mortality Rate (2017-2021)	Exposure PM2.5 on average 10 Years (2011-2021)				Exposure PM2.5 on average 5 Years (2017-2021)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
PM2.5 (Jan-Dec)	0.392*** -0.125	0.275* -0.138			0.427*** -0.134	0.214 -0.181		
PM2.5 outside haze season (May-Dec)			0.331** -0.139	0.116 -0.147			0.378** -0.156	0.067 -0.191
PM2.5 Haze (During Jan-Apr different from May-Dec)			0.175** -0.076	0.186** -0.084			0.180** -0.078	0.104 -0.111
Control variables								
Tobacco consumption rates		✓		✓		✓		✓
Household income		✓		✓		✓		✓
Air pollutants						✓		✓
Observations	65	65	61	61	65	65	61	61
R-squared	0.142	0.183	0.175	0.233	0.156	0.273	0.184	0.31

1. On average PM2.5 correlated with the mortality rate from lung cancer for **10 years exposure windows**.
2. For every one microgram per cubic meter air increase in PM2.5 ($\mu\text{g}/\text{m}^3$), the probability of lung cancer mortality rate increases by **0.275 percentage point**.
3. However, we cannot conclude the linked of PM2.5 with lung cancer mortality rate in short-term or 5 years exposure in case of Thailand.

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Surveys on tobacco consumption for 20 and 10 years of exposure were conducted in 2011 and for 5 years in 2018.

Results

Table 3: The impact of AOD for 20- 10- and 5-years exposure on lung cancer mortality

Y = Average Lung cancer Mortality Rate (2017-2021)	Exposure AOD on average 20 Years (2001-2021)				Exposure AOD on average 10 Years (2011-2021)				Exposure AOD on average 5 Years (2017-2021)			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
AOD (Jan-Dec)	0.036*** -0.003	0.015*** -0.004			0.038*** -0.003	0.008 -0.005			0.041*** -0.004	0.003 -0.011		
AOD outside haze season (May-Dec)			-0.019*** -0.006	-0.024*** -0.006			-0.014** -0.007	-0.030*** -0.007			-0.016** -0.007	-0.018 -0.011
AOD Haze (During Jan-Apr different from May-Dec)			0.052*** -0.004	0.047*** -0.004			0.043*** -0.004	0.036*** -0.005			0.044*** -0.004	0.036*** -0.007
Control variables												
Tobacco consumption rate		✓		✓		✓		✓		✓		✓
Household income		✓		✓		✓		✓		✓		✓
Air pollutants										✓		✓
Observations	928	928	928	928	928	928	928	928	928	928	928	928
R-squared	0.093	0.148	0.221	0.245	0.091	0.165	0.175	0.225	0.096	0.225	0.194	0.255

1. On average AOD correlated with the mortality rate from lung cancer for **20 years exposure windows**.
2. According to the twelfth, sixteenth, and twentieth regression models, the season of haze was the source of the positive correlation between AOD and lung cancer mortality.

Conclusion

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Is there a connection between lung cancer and the effects of transboundary haze in northern Southeast Asia?

- The health impact analysis demonstrates a significant correlation between long-term exposure PM2.5 and lung cancer mortality.
- The season of haze was the source of the positive correlation between PM2.5 and lung cancer mortality.
- However, we cannot conclude the linked of PM2.5 with lung cancer mortality rate in short-term or 5 years exposure in case of Thailand.